



Chemical Thermodynamics

Energy changes in chemical reactions and physical processes

1. INTRODUCTION

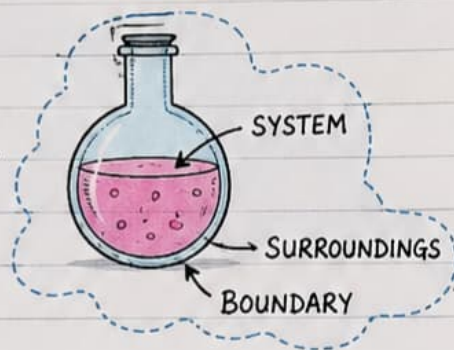
- Thermodynamics is the study of **energy changes** that occur during **chemical** and **physical** processes.
- It helps us to predict the **feasibility**, **direction** and **extent** of a process.
- It is divided into two types:
 - Thermochemistry** → deals with **heat changes**.
 - Thermodynamics** → deals with **energy changes** in general.

KEY IDEA

Energy can neither be **created** nor be **destroyed**, it can only be **transformed** from one form to another.

2. SYSTEM AND SURROUNDINGS

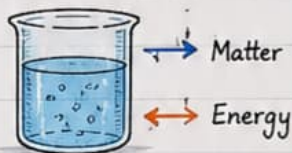
- System** → The part of universe in which the change occurs.
- Surroundings** → Everything outside the system.
- Boundary** → The surface that separates the system from its surroundings.



3. TYPES OF SYSTEM

(i) OPEN SYSTEM

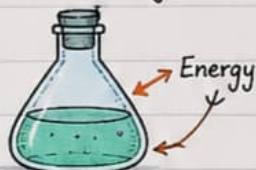
Matter and **energy** both are exchanged with surroundings.



Example : Boiling water in an open beaker

(ii) CLOSED SYSTEM

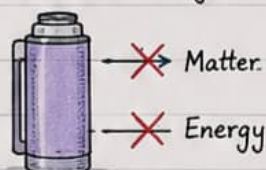
Only **energy** is exchanged, **matter** is not exchanged.



Example : Water in a closed flask

(iii) ISOLATED SYSTEM

Neither **matter** nor **energy** is exchanged with surroundings.



Example : Thermos flask

4. THERMODYNAMIC VARIABLES

- State functions** whose values depend only on the state of the system, not on the path.
- Important thermodynamic variables :

Pressure
(P)

Volume
(V)

Temperature
(T)

Internal energy
(U)

Enthalpy
(H)

Entropy
(S)

Remember → All spontaneous changes lead to a **decrease** in free energy of the system.

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5. THERMODYNAMIC FUNCTIONS



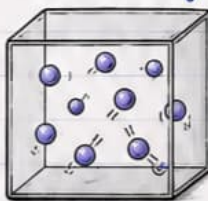
These are **state functions** (depend only on the initial and final state, not on the path).

(ii) INTERNAL ENERGY (U)

- Total energy possessed by the system due to the **motion** and **interaction** of its particles.
- Change in internal energy :

$$\Delta U = U_{\text{final}} - U_{\text{initial}}$$

Particles in a gas



Particles are always in constant motion
→ possess energy.

(ii) ENTHALPY (H)

- Total **heat content** of the system.
- For a system :

$$H = U + PV$$

(P = pressure, V = volume)

At constant pressure,
change in **enthalpy** (ΔH)
= heat absorbed or released by the system.

(iii) ENTROPY (S)

- Measure of **randomness** or **disorder** in the system.
- For reversible process :

$$dS = \frac{\delta q_{\text{rev}}}{T} \quad (T = \text{absolute temperature})$$

- For isolated system :

$\Delta S > 0$ (increase in disorder, spontaneous)

$\Delta S = 0$ (reversible process)

$\Delta S < 0$ (not possible for spontaneous process)

VISUAL IDEA

Ice cube (ordered)
melts to water
(more disordered)

→ $\Delta S > 0$



6. FIRST LAW OF THERMODYNAMICS



It is the law of **conservation of energy**.

$$\Delta U = q + w$$

Change in internal energy

Heat exchanged
(+ve if absorbed)

Work done on the system
(+ve if done on the system)

SIGN CONVENTION

- Heat absorbed by the system → $q = +ve$
- Heat released by the system → $q = -ve$
- Work done on the system → $w = +ve$
- Work done by the system → $w = -ve$



Work done on system
(+ve)

Work done by system
(-ve)

7. ENTHALPY CHANGE (ΔH)



At constant pressure, $\Delta H = q_p$ (heat exchanged)

(ii) Enthalpy of reaction (ΔH_{rxn})

Heat change when reactants are converted into products.



$$\Delta H_{\text{rxn}} = \sum H(\text{products}) - \sum H(\text{reactants})$$

(iii) Enthalpy of formation (ΔH_f°)

Enthalpy change when 1 mole of a compound is formed from its elements in their standard states.

Example :

$$\Delta H_f^\circ [\text{H}_2\text{O}(l)] = -285.8 \text{ kJ mol}^{-1}$$

(iii) Enthalpy of combustion (ΔH_c°)

Enthalpy change when 1 mole of substance is completely burned in O_2 .

$$\text{Example : } \Delta H_c^\circ [\text{CH}_4(g)] = -890.3 \text{ kJ mol}^{-1}$$

TYPES OF REACTIONS BASED ON ΔH

(a) Exothermic Reaction ($\Delta H < 0$)

Heat is released by the system.
Products have less enthalpy than reactants.
Energy level diagram :



(b) Endothermic Reaction ($\Delta H > 0$)

Heat is absorbed by the system.
Products have more enthalpy than reactants.
Energy level diagram :



Remember !

- ✓ $\Delta H < 0$
Exothermic
- ✓ $\Delta H > 0$
Endothermic
- ✓ $\Delta H = 0$
Thermoneutral



Thermodynamics helps us predict whether a process is **possible**, **feasible** and what amount of **energy** is involved.



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8. GIBBS FREE ENERGY (G) ★

- Another state function that helps us to predict the spontaneity of a process at **constant temperature and pressure**.

Defined as : $G = H - TS$

(G = Gibbs free energy, H = Enthalpy,
T = absolute temperature, S = Entropy)

Spontaneity and ΔG :

$\Delta G < 0 \rightarrow$ Process is spontaneous

$\Delta G = 0 \rightarrow$ Process is at equilibrium

$\Delta G > 0 \rightarrow$ Process is non-spontaneous

What does G mean?

★ H $\rightarrow$ total energy content

★ TS $\rightarrow$ energy not available to do useful work (randomness)

★ So, G $\rightarrow$ maximum useful energy available.

VISUAL IDEA

Think of energy as a bucket of water.



TS
(unusable, lost in randomness)

G
(usable, does useful work)

Relation with Equilibrium :

• At equilibrium, $\Delta G = 0$

• For a reaction :

$$\Delta G^\circ = -RT \ln K$$

(R = gas constant, T = temperature, K = equilibrium constant)

If $K > 1 \rightarrow \Delta G^\circ < 0$ (spontaneous)

If $K < 1 \rightarrow \Delta G^\circ > 0$ (non-spontaneous)

9. STANDARD ENTHALPY CHANGES (ΔH°) ★

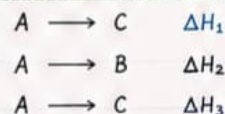
Enthalpy changes are measured under **standard conditions** (1 bar pressure, 298 K temperature).

Type	Meaning
(i) ΔH_f° Standard enthalpy of formation	Enthalpy change when 1 mole of a compound is formed from its elements in their standard states.
(ii) ΔH_c° Standard enthalpy of combustion	Enthalpy change when 1 mole of a substance is completely burned in O_2 .
(iii) ΔH_d° Standard enthalpy of decomposition	Enthalpy change when 1 mole of a compound decomposes into simpler substances.
(iv) ΔH_n° Standard enthalpy of neutralization	Enthalpy change when 1 equivalent of acid reacts with 1 equivalent of base in solution.
(v) ΔH_{sol}° Standard enthalpy of solution	Enthalpy change when 1 mole of a substance dissolves in a large amount of solvent.

HESS'S LAW ★

The total enthalpy change for a reaction is the same, whether it occurs in one step or in several steps.

Example :



Then,

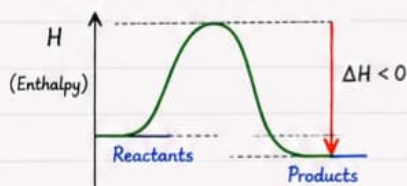
$$\Delta H_1 = \Delta H_2 + \Delta H_3$$



This helps calculate ΔH for a reaction that cannot be measured directly.

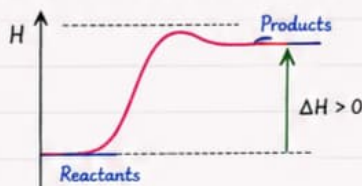
10. ENERGY DIAGRAMS (ΔH and ΔG) ★

(i) Exothermic Reaction ($\Delta H < 0$)



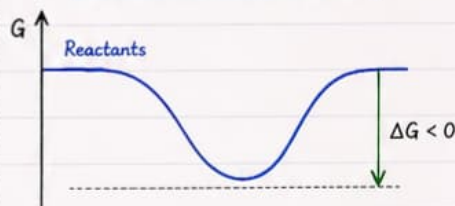
Products have less energy than reactants. Heat is released.

(ii) Endothermic Reaction ($\Delta H > 0$)



Products have more energy than reactants. Heat is absorbed.

Gibbs Free Energy Diagram



Decrease in G ($\Delta G < 0$) means the process is spontaneous.

11. APPLICATIONS OF THERMODYNAMICS ★

1. Predicting Spontaneity

Helps us to predict whether a reaction will occur or not.



2. Equilibrium

Helps in calculating equilibrium constant and predicting the position of equilibrium.

3. Industrial Processes

Used in Haber process, Contact process, and metallurgy.



4. Biological Systems

Energy changes in respiration, photosynthesis, ATP synthesis etc. follow thermodynamic principles.

5. Material Stability

Helps in understanding why some substances are stable and others react.



KEY TAKEAWAY : Thermodynamics tells us not only if a reaction can occur, but also **how much energy** is involved and **in what form**.

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12. SPONTANEITY AND EQUILIBRIUM

- A process is spontaneous in the direction in which ΔG is negative.
- At equilibrium, $\Delta G = 0$.
- For a process at constant temperature and pressure, the change in Gibbs free energy decides the feasibility.

ΔG	Nature of Process	At Equilibrium
$\Delta G < 0$	Spontaneous	Products are favoured
$\Delta G = 0$	At equilibrium	No net change
$\Delta G > 0$	Non-spontaneous	Reactants are favoured

13. EQUILIBRIUM CONSTANT (K)

- For a general reaction: $aA + bB \rightleftharpoons cC + dD$
- Equilibrium constant (K) at temperature T:

$$K = \frac{[C]^c [D]^d}{[A]^a [B]^b} \quad (\text{in concentration terms})$$

IMPORTANT POINTS

- Value of K depends only on temperature.
- Larger the value of K, greater is the extent of reaction.
- K has no unit.

- In terms of partial pressures (for gaseous reaction):

$$K_p = \frac{(P_C)^c (P_D)^d}{(P_A)^a (P_B)^b}$$

15. EFFECT OF TEMPERATURE ON SPONTANEITY

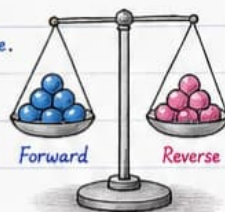
ΔH	ΔS	Effect of Temperature	Spontaneity
-	+	Favoured at all T	Always spontaneous
+	-	Not favoured at any T	Never spontaneous
-	-	Favoured at low T	Spontaneous at low T
+	+	Favoured at high T	Spontaneous at high T

17. SUMMARY CHEAT SHEET

Internal energy change	$\Delta U = q + w$
Enthalpy change	$\Delta H = \Delta U + P\Delta V$
Entropy change (reversible)	$dS = \frac{\delta q_{rev}}{T}$
Gibbs free energy	$G = H - TS$
Spontaneity	$\Delta G < 0$ (spontaneous)
Equilibrium	$\Delta G = 0$
Non-spontaneity	$\Delta G > 0$ (non-spontaneous)
Relation	$\Delta G^\circ = -RT \ln K$
Max. useful work	$w_{max} = -\Delta G$

HOW DO SYSTEMS REACH EQUILIBRIUM?

- A system changes to reduce its free energy.
- It moves in a direction where ΔG is negative.
- When ΔG becomes zero, the system reaches equilibrium.
- At equilibrium, forward and reverse processes occur at equal rates.



At equilibrium rates are equal

14. RELATION BETWEEN ΔG° AND K

$$\Delta G^\circ = -RT \ln K$$

Where,

R = universal gas constant = $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$

T = temperature in K

K = equilibrium constant

CONCLUSIONS

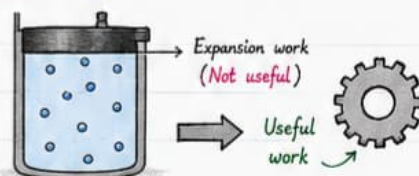
- If $K > 1 \rightarrow \Delta G^\circ$ is negative $\rightarrow$ reaction is spontaneous
- If $K = 1 \rightarrow \Delta G^\circ = 0 \rightarrow$ reaction is at equilibrium
- If $K < 1 \rightarrow \Delta G^\circ$ is positive $\rightarrow$ reaction is non-spontaneous

16. MAXIMUM WORK AND ΔG

- The maximum useful (non-expansion) work obtainable from a process at constant T and P is equal to the decrease in Gibbs free energy.

$$w_{max} = -\Delta G \quad (\text{at constant T and P})$$

Only non-expansion work is useful.



18. KEY TAKEAWAYS

- Chemical thermodynamics helps us understand energy transformations in chemical processes.
- Nature decides spontaneity by changes in Gibbs free energy (ΔG).
- At equilibrium, $\Delta G = 0$ and there is no net change.
- Equilibrium constant (K) tells us the extent of a reaction.
- Thermodynamics connects heat, work, energy and direction of chemical reactions.



Remember

Energy can neither be created nor destroyed, but it can be transformed from one form to another.

